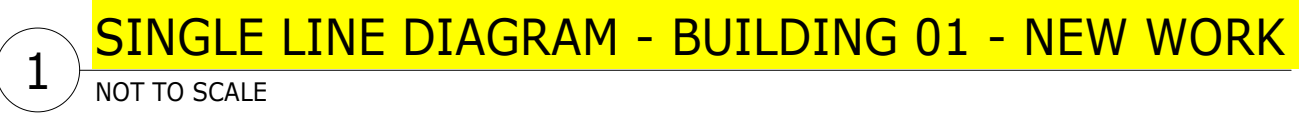


ELECTRICAL DISTRIBUTION SYSTEM (M1-B01. ES-F715)



GENERAL NOTES

1. ALL WORK SHOWN ON ELECTRICAL DRAWINGS SHALL BE BY A LICENSED ELECTRICAL CONTRACTOR UNLESS OTHERWISE NOTED.
2. ALL INSTALLATIONS AND WORK SHALL BE DONE PER NEC AND LOCAL APPLICABLE CODES AND REGULATIONS.
3. FOR GENERAL NOTES, LEGENDS, AND ABBREVIATIONS, REFER TO DRAWING M1-F600-000
4. COORDINATE WORK WITH ARCHITECTURAL, MECHANICAL, STRUCTURAL, AND ALL OTHER TRADES.

KEYNOTES

1. UTILIZE EXISTING MAIN 300A PANEL AND REPLACE BRANCH BREAKERS AS NEEDED AND AS SHOWN ON THIS DRAWING. BRANCH BREAKERS WITH AN ASTERISK ARE NEW.
2. EXISTING PANEL SHALL BE PROVIDED FOR REUSE. PROVIDE NEW LABELING AND PANEL DIRECTORY AS SHOWN ON THIS DRAWING AND PANEL SCHEDULE.
3. PROVIDE PANELBOARDS WITH NEMA 1 ENCLOSURE AND AS SHOWN ON THIS DRAWING, SPECIFICATION AND PANEL SCHEDULES.
4. PROVIDE DRY TYPE TRANSFORMERS AS SHOWN ON THIS DRAWING AND PER SPECIFICATION.
5. RESIDUAL CURRENT MONITORING (RCM) SHALL CONSIST OF AN ENCLOSURE MOUNTED NEAR EXISTING PANEL. ENCLOSURE MOUNTED DEVICES SHALL BE BENDER MODEL NO. RCM460-02 120VAC SUPPLY, 12 CHANNEL WITH ONE (1) CORE BALANCE CT (CBCT) FOR EACH CIRCUIT.
6. PER NEC, PROVIDE PROPER GROUNDING FOR PANELS AND BONDING FOR TRANSFORMERS.
7. PROVIDE NEW METERING AND ASSOCIATED ENCLOSURE, SOCKETS AND WIRING/CONDUITS.

LEGEND

* NEW BRANCH BREAKERS TO BE PROVIDED BY EC.

LEGEND OF 600V FEEDER SIZES - COPPER CONDUCTORS					
FEEDER SYMBOL	CONDUCTORS (3 PHASE, 3 WIRE) WITH GROUND	RAYCEAY SIZE	CONDUCTORS (3 PHASE, 4 WIRE) WITH GROUND	RAYCEAY SIZE	NOMINAL AMPERE RATING
1	3/C #12 W/#12 GND.	3/4"	4/C #12 W/#12 GND.	3/4"	20
2					
3					
4	3/C #10 W/#10 GND.	3/4"	4/C #10 W/#10 GND.	3/4"	30
5					
6					
7	3/C #8 W/#8 GND.	3/4"	4/C #8 W/#8 GND.	3/4"	40
8					
9					
10	3/C #4 W/#8 GND.	1"	4/C #4 W/#8 GND.	1-1/2"	70
11					
12					
13	3/C #2 W/#6 GND.	1"	4/C #2 W/#6 GND.	1-1/2"	100
14					
15					
16	3/C #1 W/#6 GND.	1-1/4"	4/C #1 W/#6 GND.	1-1/2"	125
17					
18					
19	3/C #1/0 W/#6 GND.	1-1/4"	4/C #1/0 W/#6 GND.	1-1/2"	150
20					
21					
22	3/C #2/0 W/#6 GND.	1-1/2"	4/C #2/0 W/#6 GND.	2"	175
23					
24					
25	3/C #3/0 W/#4 GND.	2"	4/C #3/0 W/#4 GND.	2"	200
26					
27					
28	3/C #4/0 W/#4 GND.	2"	4/C #4/0 W/#4 GND.	2"	225
29					
30					
31	3/C 250 kcmil W/#4 GND.	2"	4/C 250 kcmil W/#4 GND.	2-1/2"	250
32					
33					
34	3/C 350 kcmil W/#3 GND.	2-1/2"	4/C 350 kcmil W/#3 GND.	3"	300
35					
36					
37	3/C 500 kcmil W/#2 GND.	3"	4/C 500 kcmil W/#2 GND.	3"	350
38					
39					
40	3/C 600 kcmil W/#2 GND.	3"	4/C 600 kcmil W/#2 GND.	3-1/2"	400
41					
42					
43	2 SETS - 3/C 250 kcmil W/#4 GND.	(2) 2"	2 SETS - 4/C 250 kcmil W/#4 GND.	(2) 2 1/2"	500
44					
45					
46	2 SETS - 3/C 350 kcmil W/#3 GND.	(2) 2"	2 SETS - 4/C 350 kcmil W/#3 GND.	(2) 3"	600
47					
48					
49	2 SETS - 3/C 600 kcmil W/#2 GND.	(2) 3"	2 SETS - 4/C 600 kcmil W/#2 GND.	(2) 3-1/2"	800
50					
51					
52	3 SETS - 3/C 500 kcmil W/#2 GND.	(3) 3"	3 SETS - 4/C 500 kcmil W/#2 GND.	(3) 3-1/2"	1000
53					
54					
55	4 SETS - 3/C 500 kcmil W/#2 GND.	(3) 3"	4 SETS - 4/C 500 kcmil W/#2 GND.	(3) 3-1/2"	1200
56					
57					
58	4 SETS - 3/C 750 kcmil W/#1 GND.	(4) 3"	4 SETS - 4/C 750 kcmil W/#1 GND.	(4) 3-1/2"	1600
59					
60					

NOTES

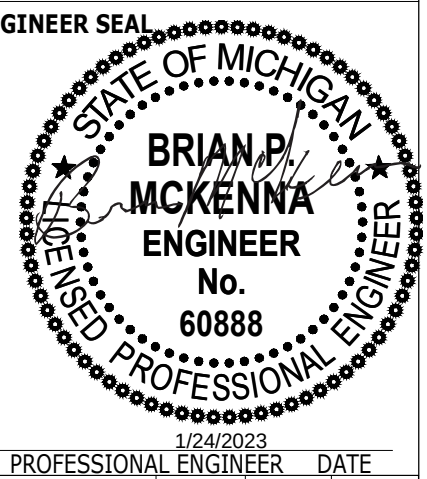
1. FEEDER SIZES ARE BASED ON 75C. THIS CONSIDERS EQUIPMENT IS LISTED AND IDENTIFIED FOR USE WITH SUCH CONNECTIONS; NFPA 70 110.14 ELECTRICAL CONNECTIONS (C) (1) (3).
2. STANDARD CABLE TYPE SHALL BE 600V, UL, TYPE TC-ER, 90 DEG. C, XHHW-2.
3. 15KV CABLE TYPE SHALL BE 3 #4/0, 15KV, SHLD, EPR, TYPE MV-105, OKFONITE CAT # 115-23-3236, W/ 1 # 1/0 BC GND.
4. A "V" IN THE FEEDER SYMBOL INDICATES THAT THE WIRE SHALL BE VFD MOTOR FEEDER TYPE.
5. VFD MOTOR FEEDER TYPE SHALL BE 600V, UL, TYPE TC-ER, COPPER TAPE SHIELD, 90 DEG. C, XLPE, W/ 3 SYMMETRICAL GROUND CONDUCTORS.

CIRCUIT LENGTH		
WIRE SIZE	120V CIRCUIT	277V CIRCUIT
#10	65 FT. TO 120 FT.	150 FT. TO 240 FT.
#8	120 FT. TO 180 FT.	ABOVE 240 FT.
#6	ABOVE 180 FT.	-

ENGINEER: DPS Group (MI), INC.
175 REGENCY WOODS, SUITE 400 CARY, NC 27518

CONFIDENTIAL

KEY PLAN

[illegible]

0	2023-JAN-25	ISSUED FOR CONSTRUCTION	KEP	NOF	DLM
REV	DATE	DESCRIPTION	DESR	CHKD	APPR
LOCATION					

 **cytiva**
1740 Whitehall Rd, Muskegon, MI 49445

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PROJECT K STAR

TITLE
ELECTRICAL
SINGLE LINE DIAGRAM - NEW WORK

PROJECT NO. D21RD011	SCALE N.T.S.	CLIENT PROJECT NO. 218492
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DRAWING NO.		REV
M1-E600-B01-300		0